

- `net.core.wmem_default` – the default buffer size used for the transmit buffer
- `net.core.rmem_max` – the maximum receive buffer used by an application
- `net.core.wmem_max` – the maximum send buffer used by an application

We can always find out the current values of these parameters using the `sysctl -a` command as follows:

```
[root@legacy ~]# sysctl -a | grep net.core
```

```
net.core.rmem_max = 131071
```

```
net.core.rmem_default = 114688
```

```
[root@legacy ~]#
```

```
[root@legacy ~]# sysctl -a | grep net.core
```

```
net.core.wmem_max = 131071
```

```
net.core.wmem_default = 114688
```

```
net.core.warnings = 1
```

```
[root@legacy ~]#
```

We'll now try to understand how to tune the size of the TCP buffer.

A high-bandwidth connection is also known as a fat pipe, and a high-latency connection is also referred to as a long pipe. For maximum throughput, we would like to transmit continuously. TCP buffers can be tuned by using the `net.ipv4.tcp_rmem` parameter for reads and `net.ipv4.tcp_wmem` parameter for writes. For each, `sysctl` holds three values: the minimum, the default and the maximum size.

For tuning the TCP buffer, we need to calculate the BDP (Bandwidth Delay Product), i.e., the length of pipe. The formula for calculating the BDP is: $Length\ of\ Pipe = Bandwidth \times Delay\ (RTT)$, i.e., $L = B \times D$. The value of BDP is in bytes.

Using `ping`, we can determine the RTT value. Let's assume that we are using a 100 Mbps NIC, and the value of time that `ping` gives us is roughly 25.999 ms.

```
So the BDP = L = 100 MiBps x 25.999 ms
              = [(100 x 1024 x 1024) x
                (0.025)] / 8 bytes
              = 327680 bytes
```

Let's look at the default values for the TCP read buffer in our system:

```
[root@legacy ~]# cat /proc/sys/net/ipv4/tcp_rmem
```

```
4096 87380 3366912
```

```
[root@legacy ~]#
```

```
[root@legacy ~]# sysctl -a | grep tcp_rmem
```

```
net.ipv4.tcp_rmem = 4096 87380 3366912
```

```
[root@legacy ~]#
```

Now I can change the values that have been calculated by BDP, by using:

```
echo "327680 327680 327680" > \
/proc/sys/net/ipv4/tcp_rmem
```

...or I can edit the `/etc/sysctl.conf` file, as shown below:

```
net.ipv4.tcp_rmem = "327680 327680 327680"
```

Remember to run `sysctl -p` after making any changes to `/etc/sysctl.conf`. Also remember to take a look at the man page for the `tc` command.

With this, I conclude this series of articles on 'Performance Tuning and Monitoring'. I do hope I have been able to share some good tips with you, and look forward to sharing with you something more exciting, soon. 

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